

Objective

To implement and demonstrate Interface Inheritance in Java, where interface I1 extends interface I2, and class C1 implements I1.

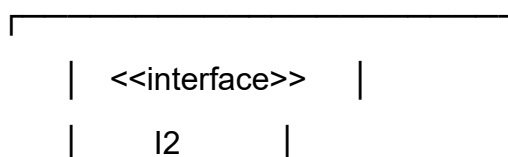
Steps for Execution

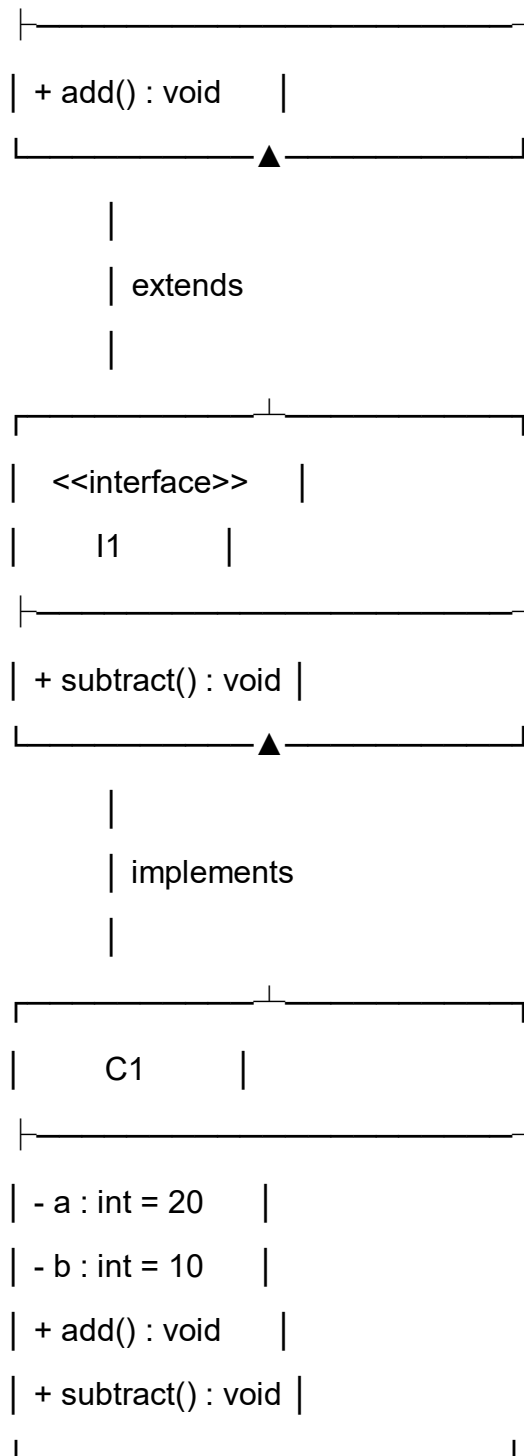
- 1.Create a Java project and package named javacore.
- 2.Create a file named Main.java and enter the given code.
- 3.Define interface I2 with the add() method.
- 4.Define interface I1 extending I2 with the subtract() method.
- 5.Create class C1 implementing I1 and define both methods.
- 6.Create a C1 object and call the methods.
- 7.Compile and run the program.
- 8.Observe the output.

Algorithm

- 1.Start the program.
- 2.Define interface I2 with add().
- 3.Define interface I1 extending I2 with subtract().
- 4.Create class C1 implementing I1.
- 5.Implement add() and subtract().
- 6.Create a C1 object.
- 7.Call both methods.
- 8.Display the output.
- 9.Stop the program.

UML CLASS DIAGRAM:





Relationships:

- I1 extends I2
- C1 implements I1
- C1 implements both `add()` and `subtract()`.

CODE:

```
package javacore;

interface I2 {
    void add();
}

interface I1 extends I2 {
    void subtract();
}

class C1 implements I1 {

    int a = 20;
    int b = 10;

    @Override
    public void add() {
        System.out.println("Addition = " + (a + b));
    }

    @Override
    public void subtract() {
        System.out.println("Subtraction = " + (a - b));
    }
}

public class Main {

    public static void main(String[] args) {

        C1 obj = new C1();
```

```
        obj.add();  
        obj.subtract();  
    }  
}
```

OUTPUT:

Addition = 30

Subtraction = 10

Result:

The program successfully demonstrates Interface Inheritance in Java. Interface I1 extends I2, and class C1 implements I1, thereby implementing both add() and subtract() methods.